# ■ FraudLens AI: Viva Defense Q&A; Guide

**Top 10 High-Yield Examination Questions & Mathematically Grounded Answers**
Candidate: Manish Patil | Project: FraudLens AI (Machine Learning PBL 100 Marks)
**Live App:** https://fraudlensai1.streamlit.app/ | **GitHub:** https://github.com/Manish-Patil33/FraudLens-AI-

---

## Q1: Why did you prioritize Precision-Recall AUC (PR-AUC) over Accuracy and ROC-AUC?

In rare fraud (0.5% prevalence), **Accuracy** is deceptive because predicting 100% legitimate transactions yields 99.5% accuracy while catching $0 of fraud.
**ROC-AUC** plots TPR vs FPR. Because True Negatives (TN) is massive, a large increase in False Positives barely shifts the denominator of FPR = FP / (FP + TN), masking false alerts.
**PR-AUC** plots Precision (TP / (TP + FP)) vs Recall (TP / (TP + FN)). Both metrics focus strictly on the minority fraud class. A single false alert immediately drops precision, making PR-AUC the gold standard for rare-event fraud.

## Q2: How did you ensure there was zero data leakage in your pipeline?

1. **Split Before Transform:** We executed Stratified Splitting into Train (80%) and Test (20%) strictly prior to any feature transformations or scaling.
2. **Fitted Transformers on Train Only:** RobustScaler computed medians and IQR solely from X_train; X_test was only transformed.
3. **Runtime Features Only:** Engineered features (balance errors, ratios) only use attributes known at transaction initiation time.

## Q3: What is the core rationale for combining Supervised Classification with Anomaly Detection?

**Supervised Models** (Random Forest, XGBoost) excel at catching known fraud patterns present in training labels. However, they fail against zero-day exploits.
**Unsupervised Anomaly Detection** (Isolation Forest) models normal transaction density without labels and flags statistical outliers.
Our **Dual-Engine Fusion** (Risk = 0.7*P_clf + 0.3*S_ano) combines high precision on known attacks with an active safety net for novel attack vectors.

## Q4: How does Isolation Forest mathematically isolate anomalies?

Isolation Forest exploits the property that anomalies are 'few and different.' It recursively creates random splits in the feature space.
Normal data points in dense clusters require deep tree path lengths to isolate. Outliers sit in sparse regions and are isolated near the root with very short path lengths.
The anomaly score is s(x, n) = 2^(-E(h(x)) / c(n)), where h(x) is path length and c(n) is the average path length of an unsuccessful search in a BST. Scores near 1 denote anomalies.

## Q5: Which engineered features proved most important and why?

**1. error_balance_orig = (newbalanceOrig + amount) - oldbalanceOrg:** In legitimate transfers, origin balance drops by exact amount. In fraud, attackers siphon funds or manipulate balances, leading to massive error discrepancies.
**2. orig_account_drained:** Fraudsters rarely leave residual funds; they drain 100% of the balance.
Together, these two features accounted for over 60% of total model feature importance.

## Q6: Why did Random Forest outperform Logistic Regression and XGBoost?

• **Logistic Regression** produced 7 false alarms (precision 89.39%) because linear hyperplanes struggle with non-linear combinations of balance errors and night hour flags.
• **XGBoost** was near-perfect (PR-AUC 0.9997, 1 false alert).
• **Random Forest** with balanced subsampling achieved 1.0000 PR-AUC, 100% recall (60/60 fraud cases caught), and 0 false positives, as ensemble averaging effectively mitigated tree variance.

## Q7: Explain your 3-Tier Decision Routing Strategy and its business justification.

Outright blocking uncertain transactions frustrates legitimate cardholders, causing customer churn.
• **Score < 35 (APPROVE):** 98%+ of normal transactions proceed frictionlessly.
• **35 <= Score < 75 (CHALLENGE MFA):** Triggers instantaneous biometric/SMS OTP verification. This halts automated bot liquidations without rejecting legitimate cardholders.
• **Score >= 75 (AUTO-BLOCK):** High-risk attacks are immediately blocked and queued for fraud investigator review.

## Q8: How did you calculate the Financial Cost-Benefit and ROI?

We modeled a concrete banking balance sheet:
• **Fraud Prevented:** Sum of True Positive transaction amounts ($1,567,200.22 on test set).
• **Fraud Leaked:** Sum of False Negative amounts ($0.00).
• **Operational Overhead:** (TP * $25 investigation cost) + (FP * $35 customer friction cost) = $1,500.00.
• **Net Savings:** $1,567,200.22 - $1,500.00 = $1,565,700.22.
• **ROI:** (Net Savings / Operational Overhead) * 100 = 104,380.0%.

## Q9: What is the inference latency of FraudLens AI?

Inference takes an average of **12.4 milliseconds** per transaction on standard CPU architecture, well within the standard real-time payment gateway requirement of < 50 milliseconds.

## Q10: What are the key limitations and future enhancements for this system?

• **Limitation:** Static feature distributions may drift over years (concept drift).
• **Future Enhancements:** Incorporating Graph Neural Networks (GNNs) to detect distributed money mule syndicates, and integrating streaming feature stores (e.g. Feast/Hopsworks) for real-time velocity aggregates across millions of daily events.